

Bondal - Orlov reconstruction theorem

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Ex. (Mukai '81) A abelian var., $\check{A}$ dual abelian var.

$$D^{b\text{coh}}(A) \simeq D^{b\text{coh}}(\check{A})$$

Thm X, Y 3d CY, $X \stackrel{\text{birat'l}}{\sim} Y$, $D^{b\text{coh}}(X) \simeq D^{b\text{coh}}(Y)$.

$$xy = zw \text{ in } \mathbb{A}^4$$

blowup (x, z)

(x, w)

Th. (Balmer) $(D^{b\text{coh}}(X), \otimes) \xrightarrow[\text{Reconstruct}]{|X| \text{ is topologically noetherian}} X$

X/k smooth, projective, ω_X ample or antiample, can reconstruct X from $D^{b\text{coh}}(X)$.

Th. 1 (Bondal - Orlov) Let X be a smooth irred. projective var. over k , w/ ω_X ample or antiample. If $D^{b\text{coh}}(X)$ is equiv. as a $\triangle$ ~~graded~~ lat. to $D^{b\text{coh}}(X')$ for X' smooth proper var. $/k$, then $X \simeq X'$.

Th. 2 (Bondal - Orlov) X smooth irred. projective $/k$, ω_X ample or antiample, then the group of isom. classes of autoequivalences of $D^{b\text{coh}}(X)$ is gen. by $\text{Aut}(X)$, twists by line bundles & grading shifts.

Preliminaries - k any field.

- only consider k -linear additive cats

- cats will be graded: $T_0: D \rightarrow D$ ($n T: D \rightarrow D$)

$$P[n] = T^n P$$

Triangulated cat.: a graded cat w distinguished exact Δ .

Def'n A functor $F: D \rightarrow D'$ is graded if it commutes w grading shifts, fixed nat'l isom $t_F: F \circ T \xrightarrow{\sim} T \circ F$

A nat'l transformation M between graded functors F, G is graded if

$$F \circ T \xrightarrow{t_F} T \circ F$$

$$MT \Big| \quad \cong \quad \Big| TM$$

$$G \circ T \xrightarrow{t_G} T \circ G$$

A graded functor F is exact if it takes exact Δ to exact Δ .

A morphism between exact functors means graded nat'l transf.

Def Let D be k -linear additive cat w finite dim. Hom spaces.

A covariant functor $S: D \rightarrow D$ is called a Serre functor if it is an equiv. there are given bifunctorial isoms

$$\begin{array}{ccccc} \varphi_{A,B}: \text{Hom}_D(A,B) & \xrightarrow{\sim} & \text{Hom}_D(B, S(A))^* & \text{and s.t. the following} \\ \text{diagram commutes.} & \text{Hom}_D(A,B) & \xrightarrow{\varphi_{A,B}} & \text{Hom}_D(B, SA)^* & \\ & \downarrow & \cong & \uparrow & \\ & \text{Hom}_D(SA, SB) & \xrightarrow{\varphi_{SA,SB}} & \text{Hom}_D(SB, S^2 A)^* & \end{array}$$

Prop i) Any automorphism $\Phi: D \rightarrow D$ commutes w/ a Serre functor.

ii) Any Serre functor in a graded cat. is graded.

iii) A Serre functor in a Δ cat is exact

iv) Any two Serre functors are conn'd by a canonical graded functorial isom, which commutes w/ $\phi_{A,B}$ from the defn of a Serre functor.

§2. Reconstruction.

X smooth proper alg var. / k , $n = \dim X$

- $\omega_X[n]$ is the Serre functor on $D^b(\text{coh}(X))$ (Grothendieck-Serre duality)

$x \in X$, $k(x)$ residue field of x

$$\text{Hom}^i(P, Q) = \text{Ext}^i(P, Q) = \text{Hom}(P, Q[i])$$

Def An object P in D is called a point object of codim s if

i) $S_D(P) \simeq P[s]$

ii) $\text{Hom}^{<0}(P, P) = 0$

iii) $\text{Hom}^0(P, P)$ is a finite field ext'n of k , denoted by $k(P)$.

Prop X smooth proper alg var. / k , $\dim X = n$, ω_X or ω_X^{-1} ample.

then $P \in D^b(\text{coh}(X))$ is a point object iff $P \simeq \mathcal{O}_x[r]$ for x closed pt of X .

(All points codim n)

Prop. P is a point object in $D^b \text{Coh}(X)$. $\mathcal{H}^i =$ cohomology sheaves

$$P \otimes \omega_X[n] = S(P) \simeq P[s]$$

$$\Rightarrow P \otimes \omega_X^N[nN] \simeq P[sN], \quad \forall N \Leftrightarrow n=s$$

$$P \otimes \omega_X \simeq P$$

Def. An object $L \in D$ is invertible if for any point object $P \in D$, $\exists s \in \mathbb{Z}$

$$\text{s.t. } i) \text{ Hom}^s(L, P) = k(P)$$

$$ii) H_{n,i}(L, P) = 0 \text{ for } i \neq s$$

Prop. X smooth proper irred. alg. var. / k . Assume that all point objects

in $D^b \text{Coh}(X)$ are of the form $\mathcal{O}_x[s]$, $x \in X$, $s \in \mathbb{Z}$, then $L \in D^b \text{Coh}(X)$

is an invertible object $(\Leftrightarrow) L \simeq L[t]$ for some L l.f. and $t \in \mathbb{Z}$.

Thm. X smooth ^{proj.}, $\omega_X \simeq \omega_{X'}^{-1}$ ampl. $D^b \text{Coh}(X) \simeq D^b \text{Coh}(X')$ for some

smooth proper irred $X' \Rightarrow X \simeq X'$.